Zach Fisher

zachfisher.com ✦ linkedin.com/in/zkf ✦ me@zachfisher.com

Member of Technical Staff at OpenAI, advancing safe automation of scientific research. Previously at Google DeepMind, working on RL post-training, agent evaluation, reward modeling, and synthetic data. Former theoretical physicist and co-author of the first proof of the Quantum Null Energy Condition.

Employment

OpenAI
Member of Technical Staff — *September 2026 – Present*

- Advancing safe automation of scientific research.

Google DeepMind (formerly Google Research) — *Mountain View, CA*
Staff Software Engineer — *May 2025 – September 2026*
Senior Software Engineer — *April 2022 – May 2025*
Software Engineer — *April 2020 – April 2022*

- Developed reward-modeling and evaluation methods for reinforcement-learning post-training of large language models, including automated evaluators and feedback-driven improvement.
- Led evaluation of scientific and tool-using AI agents, including benchmark design, infrastructure, analysis, and research on simulator-based training and evaluation.
- Built synthetic-data pipelines for structured and scientific reasoning, with automated quality filtering and executable verification.
- Improved the reliability of large-scale model training and evaluation by diagnosing data, infrastructure, and reward-quality issues and building monitoring and analysis tools.
- Previously, at Google Research (pre-2024): Led attribute extraction work for Google Shopping and Maps and contributed to generative-AI shopping features announced at Google I/O 2023.
- Set technical direction across modeling, evaluation, and infrastructure projects; partnered with teams across Google product areas and supported engineers and researchers beyond the immediate team.

Perimeter Institute for Theoretical Physics — *Waterloo, ON, Canada*
Postdoctoral Researcher — *September 2017 – February 2020*

- It from Qubit fellow in quantum field theory, quantum information, and quantum gravity.
- Characterized how information escapes evaporating black holes coupled to external thermal baths, and derived entropy bounds and quantum energy conditions linking quantum information inequalities to spacetime geometry.

Education

University of California, Berkeley — *August 2012 – May 2017*
Ph.D. in Physics. Thesis: "Entropy Bounds and Entanglement." Co-formulated the Quantum Null Energy Condition and Quantum Focusing Conjecture, co-authored the QNEC's first proof, and proved a quantum generalization of the Bousso entropy bound.

Massachusetts Institute of Technology — *September 2008 – June 2012*
S.B. in Physics.

Skills

- **Research:** RL post-training, RLHF, reward and value modeling, agent evaluation, tool use, synthetic data generation, long-context architectures, mathematical reasoning, scientific AI.
- **Engineering:** Python, JAX, PyTorch, TensorFlow, C++, SQL, Apache Beam, large-scale distributed training.

Selected Publications

Machine Learning

- Gemini Team, Google (**Z. Fisher** et al.). "Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities." arXiv:2507.06261 (2025). *4,200+ citations.*
- J. Ainslie, S. Ontanon, C. Alberti, V. Cvicek, **Z. Fisher** et al. "ETC: Encoding Long and Structured Inputs in Transformers." EMNLP (2020). *520+ citations.*
- S. Ontanon, J. Ainslie, **Z. Fisher**, V. Cvicek. "Making Transformers Solve Compositional Tasks." ACL (2022). *120+ citations.*
- R. Aksitov, S. Miryoosefi, **Z. Fisher** et al. "ReST meets ReAct: Self-Improvement for Multi-Step Reasoning LLM Agent." arXiv:2312.10003 (2023). *95+ citations.*
- S. Ontanon, J. Ainslie, V. Cvicek, **Z. Fisher**. "LogicInference: A New Dataset for Teaching Logical Inference to Seq2Seq Models." arXiv:2203.15099 (2022).
- Y. Sheng, S. Gandhe, B. Kanagal, **Z. Fisher** et al. "Measuring an LLM's Proficiency at Using APIs: A Query Generation Strategy." KDD-ADS (2024).
- B. Gunel, J.B. Wendt, J. Xie, **Z. Fisher** et al. "STRUM-LLM: Attributed and Structured Contrastive Summarization." arXiv:2403.19710 (2024).

Theoretical Physics

- R. Bousso, **Z. Fisher**, S. Leichenauer, A. Wall. "Quantum Focusing Conjecture." Phys. Rev. D 93, 064044 (2016). *450+ citations.*
- R. Bousso, **Z. Fisher**, J. Koeller, S. Leichenauer, A. Wall. "Proof of the Quantum Null Energy Condition." Phys. Rev. D 93, 024017 (2016). *280+ citations; Editors' Suggestion.*
- H.Z. Chen, **Z. Fisher**, J. Hernandez, R.C. Myers, S.M. Ruan. "Information Flow in Black Hole Evaporation." JHEP 2020 (3), 152. *175+ citations.*
- R. Bousso, H. Casini, **Z. Fisher**, J. Maldacena. "Proof of a Quantum Bousso Bound." Phys. Rev. D 90, 044002 (2014). *160+ citations.*
- R. Bousso, H. Casini, **Z. Fisher**, J. Maldacena. "Entropy on a Null Surface for Interacting Quantum Field Theories and the Bousso Bound." Phys. Rev. D 91, 084030 (2015). *145+ citations.*
- H.Z. Chen, **Z. Fisher**, J. Hernandez, R.C. Myers, S.M. Ruan. "Evaporating Black Holes Coupled to a Thermal Bath." JHEP 2021 (1), 65. *115+ citations.*

Patents and Disclosures

- A. Sadr, T. Yu, D. Li, **Z. Fisher** et al. "Artificial Intelligence Generated Badges for Search." U.S. Patent Application 18/636,844 (2024).
- H. Wu, **Z. Fisher** et al. "Automated Optimization of Structured Large Language Model Prompts via Iterative Evaluation." Technical Disclosure Commons (2026).

Awards

- Product Achievement Award, Google Research (2022).
- Publication and Open Source Excellence Award, Google Research (2021).
- It from Qubit Postdoctoral Fellowship, Simons Collaboration (2017).